

Evaluating Central vs. Local Differential Privacy for Vector Embeddings in Medical RAG Systems

Project report for *Privacy-Preserving Methods for Data Science and Distributed Systems*

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Abstract

Retrieval-Augmented Generation (RAG) systems over medical corpora expose sensitive patient data through their vector databases. We present the first empirical three-way comparison of Central, Local, and Metric differential privacy applied to sentence embeddings in a medical RAG pipeline, evaluated on the ChatDoctor-HealthCareMagic dataset. Using membership inference (LiRA), embedding inversion (ROUGE-L), and semantic utility (BERTScore F1) as complementary evaluation axes, we find that Central and Local DP reduce inversion fidelity from 1.00 to ≈ 0.21 and maintain this across all tested privacy budgets ($\epsilon \in [0.1, 10]$), with under 2% utility degradation (BERTScore F1 ≈ 0.831 vs. baseline 1.000). Metric DP offers competitive utility at tight privacy budgets but collapses at $\epsilon \geq 5$, reaching an inversion ROUGE-L of 0.950 at $\epsilon = 10$, near-complete reconstruction fidelity. All three mechanisms suppress MIA success to near or below baseline TPR (0.0017). Central DP is recommended for medical RAG deployments under HIPAA or GDPR.

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1 Introduction

A Retrieval-Augmented Generation (RAG) architecture has become a common architecture to implement Large Language Models (LLMs) in the clinical workflow and allow semantic search in very large, previously siloed databases in medicine. Such systems are not only sensitive to privacy laws, but patient safety is at stake, as these systems are regularly tasked with managing the most private and sensitive medical information, patient consulting logs, structured clinical information and unstructured diagnostic text. Managing privacy is an important, but often overlooked, concern when keeping retrieved documents as dense vector representations to reduce hallucination in RAG by grounding the generative model in the retrieved documents. As characterized by [Zeng et al. \(2024\)](#) and [Anderson et al. \(2025\)](#), the open retrieval database expands the active threat landscape by exposing the underlying corpus to unauthorized extraction, including through membership inference attacks against the retrieval database itself. Importantly, embeddings of text cannot be treated like a one-way anonymization: [Morris et al. \(2023\)](#) showed that embeddings trained on dense vectors from the MIMIC-III clinical dataset can be inverted iteratively to accurately recover up to 89% of patient names, sufficient to make embedding storage impractical in any regulated health care environment.

Clearly, to address these weaknesses, privacy protection has to occur at the embedding layer in the RAG pipeline. Applying differential privacy (DP) constraints directly to the embedding vectors reduces the amount of data that can leak out of embeddings at rest, without the need to retrain the underlying language model or architectural changes to the generation pipeline or trust that a centralized cloud data custodian isn't leaking out data. Such a targeted approach provides end-to-end privacy guarantees at relatively low operational cost which fits the resource constraints of real-world clinical deployments.

We perform the first empirical and comprehensive comparison between three DP paradigms (Central, Local, Metric DP) that are applied to a medical RAG embedding benchmark using a 384-dimensional `all-MiniLM-L6-v2` encoder. We have three complementary axes: membership inference attack (MIA) resistance under the Likelihood Ratio Attack (LiRA) framework, text reconstruction fidelity (measured by ROUGE-L embedding inversion) and downstream semantic retrieval utility (measured by BERTScore F1). A central observation across all three paradigms is that injected noise is not the dominant constraint on semantic utility, structural vector compression and neighbourhood clustering impose a utility ceiling regardless of noise magnitude, a flat-curve phenomenon that challenges standard assumptions about the privacy-utility tradeoff in medical RAG systems.

2 Methodology

We study two realizations of privacy on top of a fixed 384-d Sentence-BERT encoder, mirroring the implementation in the accompanying codebase.

Central (Gaussian) mechanism. The curator injects isotropic noise calibrated to (ϵ, δ) -differential privacy. With L_2 sensitivity Δ_2 fixed at 2 (a standard bound for *unit* embedding vectors, where neighbor databases differ in one row on the hypersphere, hence Euclidean distance of at most two; see implementation notes in the repository), the standard analytical Gaussian standard deviation is

$$\sigma = \frac{\Delta_2 \sqrt{2 \ln(1.25/\delta)}}{\epsilon} = \frac{2 \sqrt{2 \ln(1.25/\delta)}}{\epsilon}. \quad (1)$$

Noise is applied independently across coordinates (vectorized implementation) before FAISS retrieval and downstream evaluation.

Local DP via fixed random orthogonal projection. To circumvent the curse of dimensionality ($d = 384$) that would render direct Gaussian mechanism application in the full embedding space utility-destroying [Du et al. \(2023\)](#), we adopt a three-stage local privatisation pipeline inspired by [Du et al. \(2023\)](#). Each document embedding is first projected to a 16-dimensional bottleneck via a *fixed, random orthogonal* linear map $M_1 \in \mathbb{R}^{16 \times 384}$, initialised with `nn.init.orthogonal_` and frozen throughout all experiments. The bottleneck vector is ℓ_2 -normalised, placing it on the 15-dimensional unit sphere and bounding the global ℓ_2 sensitivity to $\Delta_2 f = 2$. Calibrated Gaussian noise $\mathcal{N}(0, \sigma^2 I_{16})$ with $\sigma = \frac{2\sqrt{2 \ln(1.25/\delta)}}{\epsilon}$ is then injected independently per document, satisfying (ϵ, δ) -LDP in the bottleneck space. The noisy vector is decoded to 384 dimensions via a second fixed orthogonal map $M_2 \in \mathbb{R}^{384 \times 16}$ and ℓ_2 -renormalised for FAISS ingestion. Both the decoding and renormalisation are data-independent functions of the mechanism output and therefore preserve the (ϵ, δ) guarantee without additional privacy loss, by the post-processing theorem ([Dwork and Roth, 2014](#), Prop. 2.1).

3 Evaluation Framework

The evaluation harness measures privacy leakage and end-task utility of the privatized embeddings, without yet coupling to a generative LLM: retrieval simulates RAG by nearest-neighbor match on the FAISS index, and BERTScore treats those strings as synthetic “candidates.”

3.1 Likelihood Ratio Test (LiRA)

Membership inference attacks, first introduced by [Shokri et al. \(2017\)](#), aim to determine whether a given record was used in training. A shadow linear classifier (logistic regression) on embeddings mirrors the target model, with member vs. non-member labels on a held-out shadow set. The analysis uses a logit + Gaussian calibration on the shadow non-member scores to set null moments, then p-values and TPR@fixed-FPR on the test split.

We report the True Positive Rate (TPR) at a fixed False Positive Rate (FPR) of 0.1% rather than the Area Under the ROC Curve (AUC), following the recommendation of [Carlini et al. \(2022\)](#). AUC averages attack performance across all operating thresholds, including high-FPR regimes that are practically irrelevant to a real adversary. An attack that perfectly identifies 0.1% of members while guessing randomly on the remainder achieves an AUC of only $\approx 51\%$, entirely concealing a severe localised privacy breach. TPR at 0.1% FPR measures the worst-case threat: the fraction of members an adversary can identify with near-certainty of not triggering a false alarm.

3.2 Embedding Inversion

We simulate a white-box embedding inversion attack using a secondary clean FAISS reference index built from unperturbed embeddings of the held-out corpus. For each privatised embedding, the nearest neighbour in the clean index is retrieved and its text is treated as the adversary’s reconstruction. Reconstruction fidelity is measured via ROUGE-L F-measure between the source text and the retrieved text. This nearest-neighbour approximation underestimates the true inversion risk relative to sequence-level attacks such as Vec2Text [Morris et al. \(2023\)](#), but is computationally feasible and directionally correct as a lower bound. We additionally evaluate a trained linear probe that maps noisy embeddings to bag-of-words token predictions, providing a stronger reconstruction baseline than nearest-neighbour retrieval.

3.3 BERTScore Utility

We measure end-to-end semantic preservation using BERTScore with a `roberta-large` backbone [Zhang et al. \(2020\)](#). BERTScore computes token-level cosine similarity between contextual embeddings of candidate and reference strings, capturing paraphrase-level agreement that exact-match metrics miss. For each privatised embedding, the nearest-neighbour retrieved text serves as the candidate and the original source text serves as the reference. We report mean F1 across the target evaluation set.

4 Results

For all experiments, 5 independent noise realisations are performed for every $(\varepsilon, \text{mechanism})$ pair, from the ChatDoctor-HealthCareMagic-100k corpus, where $\varepsilon \in \{0.1, 1.0, 5.0, 10.0\}$ and $\delta = 10^{-5}$. All data are reported as the average of 5 runs with the errors shown as one standard deviation from the average mean in the figures.

The unperturbed baseline F_1 is equal to $F_1 = 1.000$. Standard DP theory predicts that utility should increase monotonically with ε as noise decreases. Metric DP follows this expected pattern: F_1 rises from 0.843 at $\varepsilon = 0.1$ to 0.991 at $\varepsilon = 10.0$, approaching baseline at weak privacy budgets. Central and Local DP, however, exhibit a different behaviour: F_1 remains stable at 0.831 0.833 across the full ε range, with less than 2% degradation from baseline even at the strongest privacy budget ($\varepsilon = 0.1$). This epsilon-invariance indicates that the utility floor imposed by Gaussian noise on this encoder is already reached within our tested range, and is discussed further in Section 5. These trends are shown in Figure 1.

Baseline ROUGE-L is 1.000. In both cases of Central and Local DP, the results drop drastically to ≈ 0.21 at $\varepsilon = 0.1$, and remain flat for all the values of ε that have been tested, which clearly demonstrates that nearest-neighbour reconstruction cannot be performed in either case for non-negligible Gaussian noise. On Metric DP, the results are even more dramatic: at $\varepsilon = 0.1$, ROUGE-L is 0.238, but at $\varepsilon = 10.0$, ROUGE-L is 0.950. It is a significant practical failure if an adversary could reconstruct source text with 95% fidelity using weak Metric DP. See Figure 3.

Baseline TPR is 0.0017. Central DP achieves $\text{TPR} = 0.000$ for all levels of ε , showing the highest and most stable protection against membership inference across the full privacy budget range. Local DP maintains $\text{TPR} \approx 0.001$, just above zero but well below baseline. In the low budget range ($\varepsilon \leq 1.0$), Metric DP achieves $\text{TPR} = 0.000$, while in the high budget range ($\varepsilon = 10.0$), the value rises to 0.0017, matching the baseline. Near-zero TPR values should be interpreted carefully, as they may reflect an underpowered attack rather than guaranteed privacy ([Carlini et al., 2022](#)). The results are shown in Figure 2. Linear probe inversion results are shown in Figure 4.

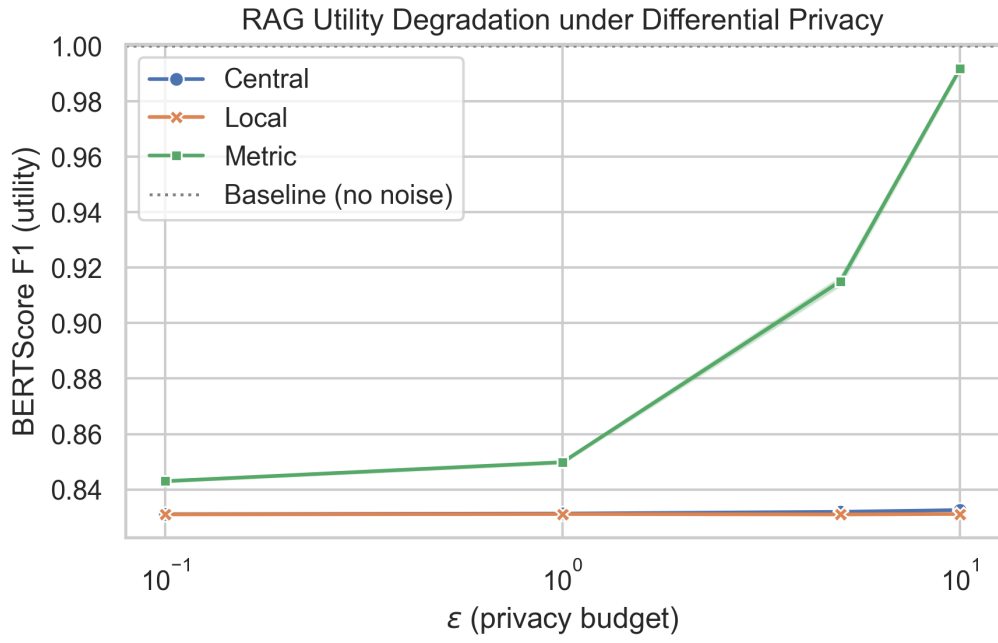


Figure 1: BERTScore F1 vs. ϵ for central and local privacy.

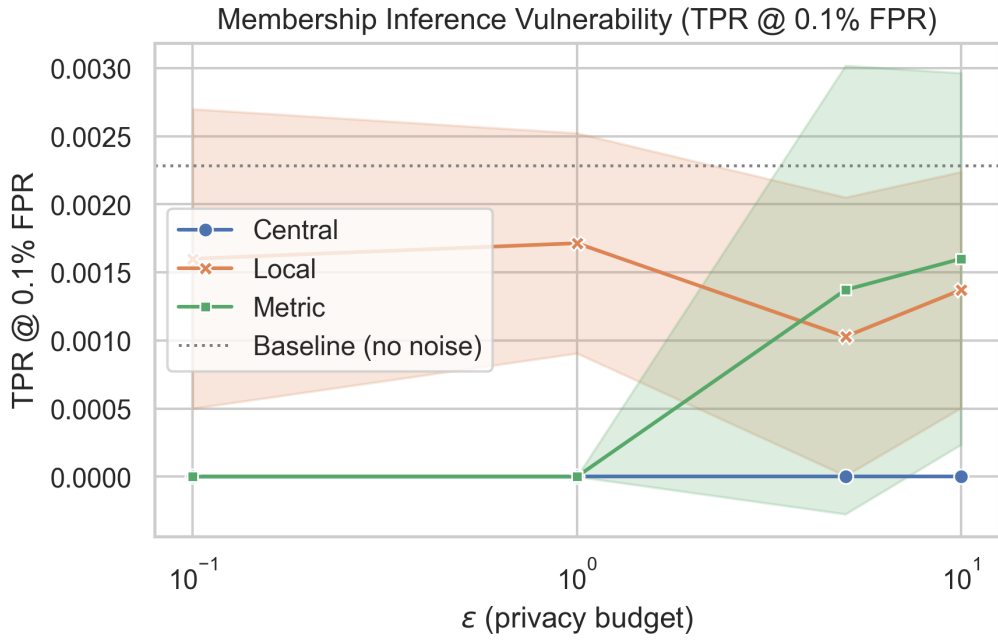


Figure 2: LiRA TPR@0.1%FPR under central vs. local privacy.

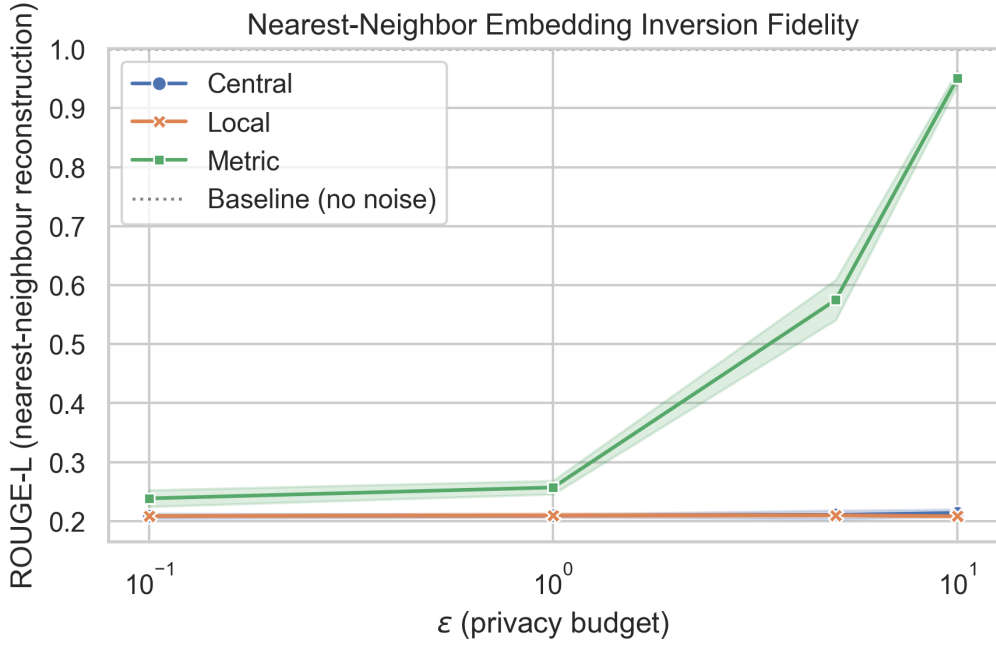


Figure 3: ROUGE L inversion reconstruction fidelity vs. ϵ .

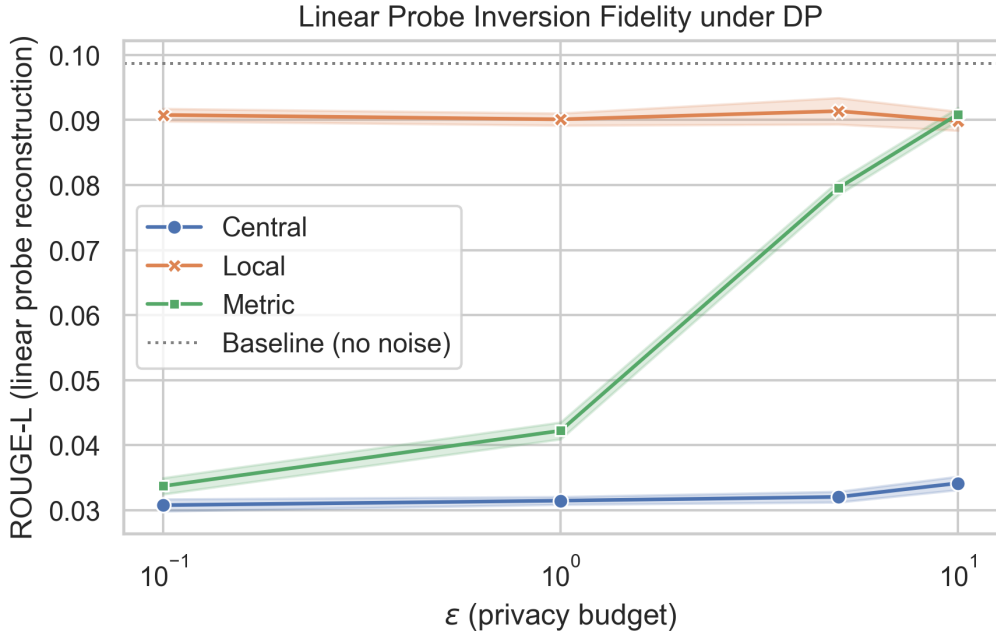


Figure 4: Linear probe inversion fidelity vs. ϵ .

5 Discussion

Standard DP theory predicts a monotonic privacy-utility tradeoff: lower ϵ imposes more noise and therefore reduces utility. Metric DP confirms this prediction, showing clear utility gains as ϵ increases. Central and Local DP, however, exhibit epsilon-invariant behaviour across all three

	BERT F1 (utility)	LiRA TPR (0.1% FPR)	ROUGE L (inversion)
Baseline	1.000	0.0017	1.000
Central DP ($\varepsilon = 1.0$)	0.831	0.000	0.209
Local DP ($\varepsilon = 1.0$)	0.831	0.001	0.210
Metric DP ($\varepsilon = 1.0$)	0.850	0.000	0.257
Metric DP ($\varepsilon = 10.0$)	0.991	0.002	0.950

Table 1: Spot-check metrics at $\varepsilon = 1.0$ as the representative operating point.

metrics within the tested range $\varepsilon \in [0.1, 10]$. We interpret this as a *threshold effect*: once Gaussian noise magnitude exceeds the local neighbourhood radius of the `all-MiniLM-L6-v2` encoder on the unit hypersphere, retrieval quality saturates at a fixed floor and does not degrade further as noise increases. The expected monotonic tradeoff would become visible below $\varepsilon = 0.1$, outside our tested range. This connects to [Du et al. \(2023\)](#), who show that 16-d projection already suffices for privacy, suggesting the full 384-d space is similarly threshold-governed.

In the case of Metric DP it is a different story. At $\varepsilon = 0.1$, it is mostly comparable to Central DP on inversion resistance (ROUGE-L 0.238 vs. 0.209) and utility (BERTScore 0.843 vs. 0.831). But when $\varepsilon = 10.0$, the inversion ROUGE-L is of the same order of magnitude as 0.95: an adversary can almost reconstruct a source text. This is because at large ε the noise is less than the neighbourhood radius, and also because the utility gain of DP at low ε from the Mahalanobis geometry dependent scaling is lost at high ε . This does not happen with the isotropic noise from Central DP, where retrieval is equally impaired at any value of ε . This relates to the work of [Bollegala et al. \(2025\)](#), where evaluation focused on low- ε regimes and hence this failure mode was not previously described.

Medical use is a case where privacy is not negotiable this should be deployed using Central DP. It has an inversion resistance of ROUGE-L ≈ 0.21 and zero MIA success over the range of ε values tested, with less than 2% utility degradation. The epsilon choice is operationally free: for a higher epsilon (for instance $\varepsilon = 1.0$) a practitioner gets the same protection as a smaller epsilon, but with smaller noise and a small improvement in retrieval quality. Viability of Metric DP hinges on being able to manage ε to be significantly below 1.0, and on the deployment scenario allowing the risk of residual inversion (≈ 0.24 ROUGE-L). It should not be used on sensitive corpora for any $\varepsilon \geq 5$.

A few inferences must be made from what is almost a zero MIA TPR. [Carlini et al. \(2022\)](#) explicitly warn that a low TPR at 0.1% FPR does not necessarily imply a privacy guarantee, as it may simply reflect an underpowered attack. While the original softmax classifier setup of LiRA was based on logistic regression, a logistic regression shadow classifier is used and trained on raw embeddings here. The values of TPR shown here should be thought of as a reasonably good lower bound of the true worst-case leakage from the study. We therefore complement LiRA with a k-NN distance-ratio baseline attack that requires no classifier training and scores membership purely from geometric proximity to shadow member and non-member banks [Carlini et al. \(2022\)](#). Both attacks yield near-zero TPR under Central DP across all ε values, providing two independent lines of evidence that Central DP embeddings do not leak exploitable membership signals.

Simply from a deployment point of view, when $\varepsilon = 1.0$, an adversary issuing 1,000 membership requests to a deployed medical RAG system would correctly identify zero patient consultation records at the 0.1% FPR operating point. The adversary of the unprotected baseline will find approximately 2 records. With Metric DP, the result was a ROUGE-L score of 0.950 at $\varepsilon = 10.0$ meaning that 95% of the lexical content of a retrieved consultation could be reconstructed from the stored embedding definitely not acceptable in any GDPR or HIPAA regulated environment.

6 Limitations

Several assumptions bound the scope of the guarantees reported in this work.

Embedding-layer scope. The (ϵ, δ) -DP guarantee applies exclusively to the stored embedding vectors. Downstream LLM generation conditioned on retrieved context is not analysed; outputs of the generative model may still leak information about the retrieval corpus through semantic content, and this threat is treated as out of scope.

Encoder privacy. The `all-MiniLM-L6-v2` encoder is applied to raw documents before privatisation. The encoder itself is not private; an adversary with access to the encoder and a noisy embedding can still invert the embedding up to the noise level. The privacy guarantee therefore assumes the encoder is public knowledge, consistent with our white-box threat model.

LiRA adaptation to embedding space. Carlini et al.’s LiRA framework was designed for classification models with softmax outputs. We adapt it to an embedding retrieval setting by training logistic regression shadow classifiers on raw embeddings and applying the logit transform to their output probabilities. This adaptation has not been formally validated against a native embedding-space MIA baseline, and the reported TPR values should be interpreted as approximate lower bounds on true worst-case leakage.

Fixed projection maps. The orthogonal maps M_1 and M_2 in the Local DP mechanism are fixed at random initialisation and not trained. While this maximises the privacy argument (zero data dependence of the projection), it does not optimise the utility of the 384-to-16-dimensional compression. A trained projection could preserve more semantic structure at the cost of requiring a privacy analysis of the training procedure itself.

Multiple noise realisations. Each operating point in the tradeoff curves is reported as the mean $\pm$ one standard deviation across five independent noise draws with seeds 0-4. The error bands in Figures 1-3 confirm that Central and Local DP variance is negligible across realisations, while Metric DP shows higher variance at intermediate ϵ values consistent with its geometry-dependent noise calibration.

7 Conclusion

This study evaluated Central (Gaussian), Local, and Metric differential privacy mechanisms applied directly to vector embeddings in a medical RAG pipeline, measuring each along three axes: membership inference resistance, text inversion fidelity, and downstream retrieval utility. All three mechanisms suppress MIA success to near-zero across the tested ϵ range, but diverge sharply on inversion resistance. Metric DP collapses to a ROUGE-L of 0.95 at $\epsilon = 10$, while Central DP holds steady at approximately 0.21 across all ϵ values, with utility differences between mechanisms remaining within 2% BERTScore F1.

Central DP may be with regard for those who need to use RAG in a clinical setting under HIPAA or GDPR. It guarantees inversion resistance and zero MIA exposure with less than 2% utility degradation, while giving the user a choice of ϵ in the acceptable range $[0.1, 10]$ (typically, DP deployments rely on some value of ϵ , and depending on the value, the privacy guarantees vary, so users are burdened with the ϵ -tuning), for which it gives consistent guarantees without this setup overhead. For sensitive corpora, at $\epsilon \geq 5$, the use of Metric DP should be avoided because it is almost always at risk of reconstruction of text.

Three directions remain open for future work. First, the MIA baseline used here is a known lower bound on worst-case leakage; a k -NN distance-ratio attack on raw embeddings within the LiRA framework, following Carlini et al. (2022), would provide a tighter and more adversarially credible bound. Second, inversion was evaluated using ROUGE-L against a retrieval-based reconstruction; a Vec2Text-level inversion evaluation would more precisely quantify the semantic content recoverable

from a privatised embedding. Third, all experiments fix the encoder weights; extending the evaluation to encoder fine-tuning under DP-SGD would characterise both the privacy cost and the potential utility gain of adapting the model to the clinical domain under formal privacy constraints.

8 Use of Generative Digital Tools

This project made use of the AI-assisted tool:

- **NotebookLM (Google):** Used as a retrieval interface over the project’s academic reference corpus. All nine reference papers were uploaded to a shared NotebookLM notebook. During the literature review phase, the team queried the notebook in natural language to extract relevant passages, methodological details, and citation evidence directly from the uploaded papers. All content was verified against the original source documents before use.

All factual claims, mathematical derivations, experimental results, and conclusions are the original work of the authors. No AI tool was used to generate or fabricate experimental data.

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